

Why the complement works:

If you receive 30% off, you pay the remaining 70%. Multiplying by 0.70 gives you the net price in one step, rather than calculating the discount amount first and then subtracting it. Both approaches give the same answer — the complement method is faster.

Part 2: Chain Discounts

Sometimes a supplier offers more than one trade discount on the same item. For example, a standard trade discount might be 20%, with an additional 10% for large orders and another 5% for preferred accounts. When multiple discounts apply, they are called **chain discounts** and written as "20/10/5."

The critical rule: **each discount applies to the result of the previous discount**, not to the original list price. This means you cannot simply add the rates together. 20/10/5 is not the same as a single 35% discount.

Chain Discount Formula:

$$\text{Net Price} = \text{List Price} \times (1-d_1) \times (1-d_2) \times (1-d_3)$$

Worked Example — Surfboard Fins:

Tidal Goods orders surfboard fins with a list price of \$64.00 per set. The supplier offers a 20/10/5 chain discount.

- Step 1: $\$64.00 \times (1 - 0.20) = \$64.00 \times 0.80 = \$51.20$
- Step 2: $\$51.20 \times (1 - 0.10) = \$51.20 \times 0.90 = \$46.08$
- Step 3: $\$46.08 \times (1 - 0.05) = \$46.08 \times 0.95 = \mathbf{\$43.78}$

Common Mistake — Adding Rates:

Adding $20 + 10 + 5 = 35\%$ would give $\$64 \times 0.65 = \41.60 — a different (lower) answer. The chain method gives \$43.78. These are not equal because each discount is applied to an already-reduced base, not the original list price.

Part 3: Cash Discounts and Payment Terms

While trade discounts reduce the price of goods, **cash discounts** reward buyers who pay their invoices early. Suppliers offer cash discounts to improve their own cash flow — getting paid sooner is